

SYNOPSIS

The Liquefied Petroleum Gas (LPG) carrier, *Gas Monarch*, collided with the sailing yacht, *Whispa*, in dense fog 6 miles ESE of Lowestoft on the evening of 16 April 2007. There were no physical injuries on either vessel. There was no damage to *Gas Monarch* and initial structural damage to *Whispa* was relatively minor. However, the damage to the yacht worsened due to progressive flooding as she was towed to Lowestoft by a lifeboat.

Gas Monarch was proceeding at full speed, in fog, when her master left the bridge in the hands of his third officer (3/O) and an able bodied seaman (AB), contrary to his own standing orders. The 3/O had been on watch for about 1 hour 40 minutes when *Whispa* appeared out of the fog on a crossing course at very short range. Evasive action by both craft was unsuccessful and the vessels collided.

Prior to the collision, the 3/O on *Gas Monarch* had detected *Whispa* by radar and had calculated that the contact would pass clear to starboard. *Gas Monarch* lost *Whispa*'s radar contact at a distance of just under 3 miles, but carried on at full speed with no sound signals in the dense fog.

Whispa was motoring on her auxiliary engine with her skipper on watch and his crew member resting below. The yacht skipper had detected *Gas Monarch* by radar and monitored the target track close to his radar heading marker, for several miles. *Whispa*'s skipper had limited knowledge of his radar's capabilities or limitations; without plotting, calculating a closest point of approach (CPA), or establishing *Gas Monarch*'s speed he concluded that the vessels were on a collision course. *Whispa* made a bold alteration to starboard when the vessels were just over a mile apart (and closing at fully 18 knots) but this action, instead of moving *Whispa* clear of *Gas Monarch*, brought the two vessels onto a collision path.

Gas Monarch's bow struck *Whispa*'s port transom and rudder, slewing the yacht round to port and pushing her clear, which allowed the vessels to pass without further contact.

The 3/O on *Gas Monarch* was in a state of shock as a result of the incident and did not slow the ship. Hearing *Whispa*'s distress call stimulated him into calling the master, who rushed to the bridge, immediately stopped the ship, and identified *Gas Monarch* to the coastguard. *Gas Monarch* stood by *Whispa* until the yacht was taken in tow by the lifeboat.

The MAIB investigation identified a number of contributing factors to the accident, including:

- A failure by both vessels to abide by collision avoidance regulations
- Deteriorated performance and accuracy of both vessels' radars
- Lack of experience by *Gas Monarch*'s third officer, compounded by lack of support from the master
- Inappropriate use of radar equipment by both vessels.

As a result of the accident, the managers of *Gas Monarch* have:

- Serviced and replaced magnetrons in both radars
- Implemented bridge team refresher training
- Reviewed and intensified its fleet audit procedures
- Recommended to her owners the replacement of electronic radar plotting aids with Automatic Radar Plotting Aids (ARPA)
- Accelerated S-VDR installation throughout its fleet.

As a result of the accident, *Whispa*'s owner has:

- Installed AIS "B" to improve detection by ships monitoring the system
- Installed a GMDSS DSC VHF radio
- Installed additional bilge pumps and bilge warning alarms.

In addition, the Oil Companies International Marine Forum (OCIMF) has:

- Added a Radar Performance Monitor guidance note to its Vessel Inspection Questionnaires
- Proposed amendments to its Tanker Management Self Assessment tool to reflect that all vessels should be fitted with ARPA as best practice.

Recommendations have been made to the Maritime and Coastguard Agency (MCA) and Comité International Radio-Maritime¹ (CIRM) regarding small commercial vessel training requirements and radar training.

¹ CIRM is an international association for companies engaged in maritime electronics whose objectives are to promote the application of electronic technology for the safety of life and efficient conduct of vessels at sea.

Figure 1



Gas Monarch

1.2 NARRATIVE

1.2.1 Background

Gas Monarch was owned by Stealth Maritime Corporation S.A., managed by Eitzen Maritime Services (EMS) Singapore and on charter to Statoil ASA.

Whispa was owned by her skipper. At the time of the accident, the yacht was being prepared for a new commercial venture. *Whispa* was so named due to her extremely good sound insulation, which cut out the majority of external noises.

1.2.2 Environmental conditions

- Dark
- Dense fog, visibility between 50 and 150m
- Light northerly wind of about 10kts
- Sea state was slight
- Two days before spring tide; southerly setting tidal stream approximately 2.0kts.

1.2.3 Events leading up to the collision

Figure 4 shows the relative tracks of *Gas Monarch*, reproduced from her Automatic Identification System (AIS), and *Whispa*, reproduced from her Global Positioning System (GPS). Frequent reference will be made to **Figure 4** throughout this report.

Gas Monarch was bound for Porto, Portugal with a cargo 2363 cubic metres (m³) of propane gas. The vessel had sailed from Immingham on the river Humber, England, at 1418 on 16 April 2007.

When *Gas Monarch* left Immingham, the prevailing visibility was approximately 3 nautical miles(nm), but this had reduced to between 100 and 150m by 2000 hours when the 3/O and an AB lookout/helmsman took over the navigational watch. No fog signals were being sounded when the relieving watch took over and, on querying this, the 3/O was advised by the chief officer not to sound fog signals unless he encountered traffic. The master of *Gas Monarch* came to the bridge several times during the first hour of the 3/O's watch and was aware of the weather conditions and the vessel's navigational status (bridge manning, speed, steering mode and sound signalling) when he went to bed at 2100.

Whispa had left Ramsgate earlier that day and was proceeding northwards towards Whitby, making good approximately 4.5kts into an adverse tidal stream. The yacht was proceeding in autopilot, under power, with no sails up. The skipper, who was on watch, was at the below deck navigation station, while the crew member was resting in the aft starboard cabin. From the below deck navigation station the skipper was able to monitor the vessel's radar display. He went to the cockpit steering position at approximately 10-minute intervals to maintain a visual lookout.

1.2.4 Pre-collision: *Gas Monarch*

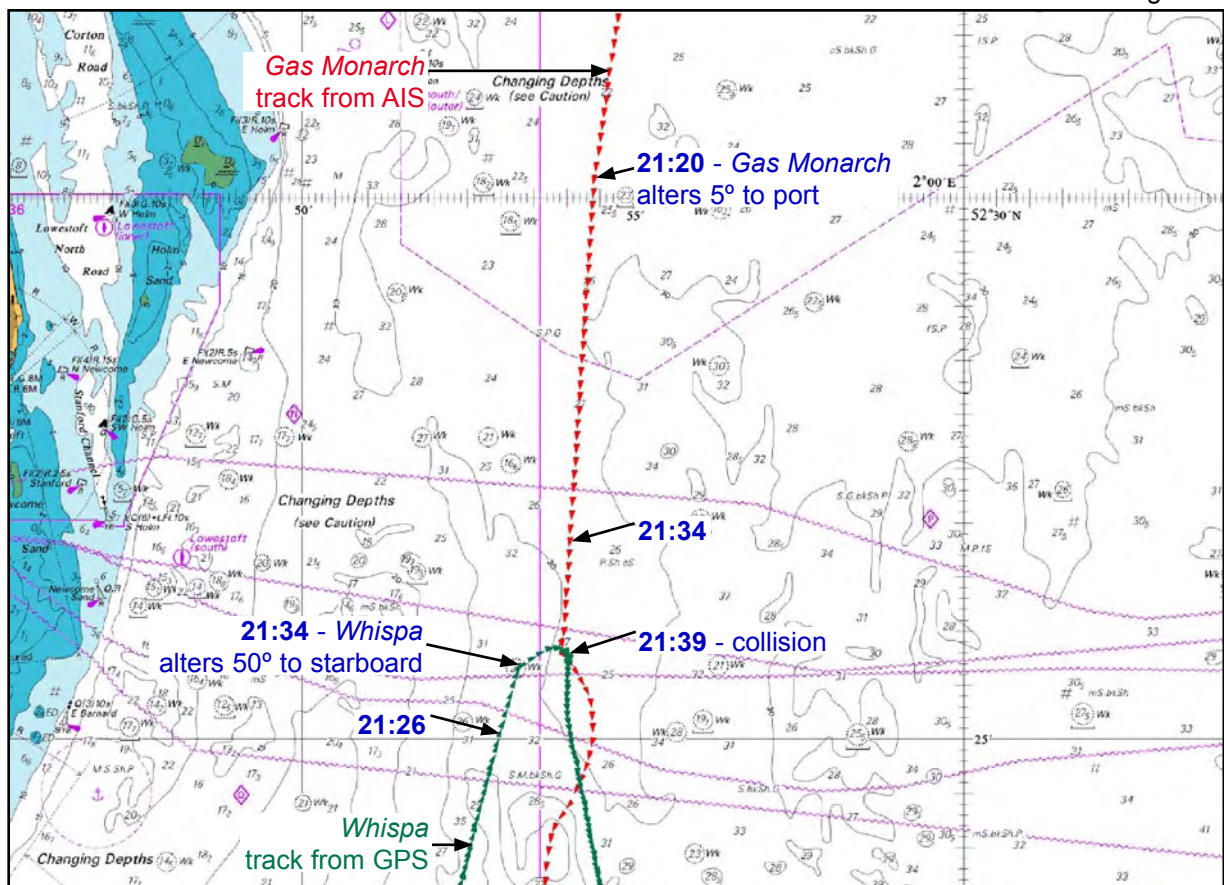
At 2118 *Gas Monarch* was proceeding on autopilot at full sea speed before a 2kt tidal stream, making good about 14.3kts, when *Whispa* was acquired as a radar contact at a range of just over 6 miles. After briefly observing the radar contact for a few minutes, when the radar contact was being displayed ahead of *Gas Monarch* at a distance of

about 5.5 miles (**Figure 4**), the 3/O altered 5° to port to create a starboard to starboard passing situation. By appraising the contact's radar trails, the 3/O estimated the resulting CPA would be approximately 1nm on *Gas Monarch's* starboard side. As the vessels closed to 3 miles (**Figure 4**), the trails of *Whispa's* radar contact indicated to the 3/O a reduced CPA of 0.7nm to starboard. When the vessels were just less than 3 miles apart, the yacht's radar contact was lost from both radar screens of *Gas Monarch*.

Once the target of *Whispa* had been lost from *Gas Monarch's* radar screen, the 3/O began to intermittently shine the beam of an Aldis lamp into the surrounding fog, a precaution to warn the approaching vessel in the event the actual CPA was less than predicted. No consideration was given to sounding the vessel's fog horn or reducing her speed.

The 3/O informed his AB that the contact was lost from the radar screens and that he should be vigilant. As *Gas Monarch* closed on *Whispa*, the 3/O continued his routine navigational duties, and some 3 minutes before the collision applied a routine fix of the ship's position to the chart. Soon after this, *Whispa's* port masthead tricolour light appeared fine on *Gas Monarch's* starboard bow. The 3/O grabbed the Aldis lamp and shone it on *Whispa* and, at the same time, instructed the AB to place the steering to manual and go hard to starboard, followed by a hard port order once the ship started to swing. The evasive action was unsuccessful and the two vessels collided at 2139.

Figure 4



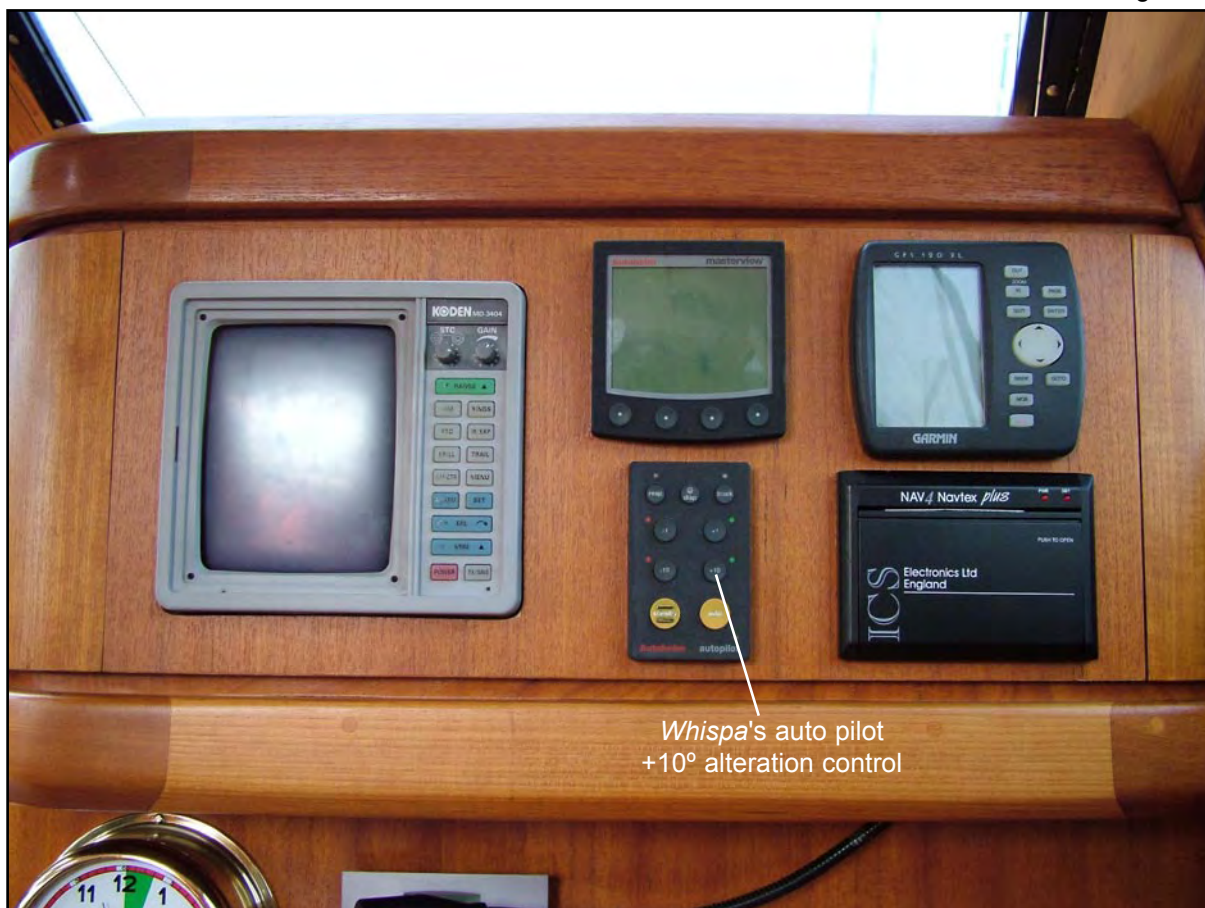
Relative tracks of *Gas Monarch* (reproduced from AIS) and of *Whispa* (reproduced from GPS)

1.2.5 Pre-collision: *Whispa*

Whispa's skipper first became aware of *Gas Monarch* as an unidentified radar target appearing at the edge of his radar's 6-mile range scale, fine on the port bow. At this stage he believed the range rings being displayed on the radar screen were set at 1-mile intervals. He did not plot the contact, but instead monitored its approach close to the radar screen's heading marker, until it was less than 3 miles away.

Whispa's skipper believed that a collision situation was unfolding, and before the contact entered his radar's alarmed guard zone, which was set at 2.5 miles, altered *Whispa*'s course 50° to starboard by pressing the autopilot 10° course alteration button 5 times in succession (**Figure 5**). The yacht skipper then reduced the range on the radar to 3nm, which he believed would display range rings at 0.5 mile intervals, and continued to monitor the contact. However, instead of tracking clear down his port side as expected, the contact appeared to still be coming towards the yacht, now from a relative bearing of 50° on the port bow. As the vessels drew closer, *Whispa*'s skipper altered a further 20° to starboard, but still the radar contact continued to close on a steady bearing. The yacht skipper rushed from the navigation station to the cockpit, put the engine on to full ahead, switched off the autopilot and put the yacht on a course 085°. As he stared into the dark, a spotlight was suddenly shone onto the yacht for some 6 to 8 seconds. The light destroyed the skipper's night vision, but as he threw the vessel hard to starboard he was able to identify a bulbous bow bearing down on him shortly before it collided with *Whispa*'s port quarter.

Figure 5



Whispa's control console

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 VISIBILITY

The 24 hour forecast transmitted by the coastguard at 1850 on 16 April 2007, gave fog patches, with visibility moderate or good but occasionally very poor. The previous forecast gave visibility as moderate or good, occasionally poor overnight. The visibility predictions were fairly accurate, and fog was not unexpected by *Gas Monarch* or *Whispa*, although it was more prolonged than anticipated.

2.3 VESSELS' RADARS

2.3.1 *Gas Monarch* radar

Gas Monarch had two type approved commercial radars, JRC type JMA7252 and 7303, neither of which had ARPA facilities. The radars did, however, have electronic tracking facilities which would have allowed operators to select a target and plot its movements across the screen at pre-set time intervals, and thus allow the operator to determine CPA. This facility was not utilised by the 3/O as he had never sailed with any type of radar other than those with ARPA; instead he used the contact's trails, floating EBL and VRM as a means of calculating CPA. The radars were interfaced with the ship's AIS receiver, and therefore displayed vessels transmitting AIS information on the radar screens. Any AIS-transmitting vessel would remain on screen even if the radar paint was lost for any reason. *Whispa*, however, did not have (nor was required to have) AIS capability.

SOLAS requires all ships of 10,000gt and upwards, and ships of 300gt and upwards built after 1 July 2002, to be fitted with ARPA. *Gas Monarch* was exempt from ARPA requirements on both age and size and, according to current regulation, will not require ARPA throughout her life. It would appear that current regulation is driven by vessel size and age rather than the type of vessel or apparent risk; in this case a petroleum gas carrier. Had ARPA facility been available, and used, it would have given a "target lost" warning when *Whispa*'s contact disappeared, and remained so until the 3/O acknowledged it. This might have had no effect on the final outcome, but it would have served to alert and highlight to the 3/O the need to take some positive action.

Several days after the collision, and at the ship manager's behest, the radars were tested using the integrated performance checking facilities, indicated in the operator's manual (**Annex 2**). This test revealed low attenuation, and therefore probable magnetron deterioration.

IMO resolution No .MSC. 64(67) requires that a means should be available to regularly evaluate radar performance. Ideally, performance level should then be recorded in some numeric or calibrated fashion to allow simple recognition of deterioration. The ship's logbook had a dedicated section for Radar Watchkeeping Performance (**Figure 16**) which, in the lead up to the accident, had been marked up in both Performance Check Results and Target Acquisition Performance as being "Good." This, of course,

is subjective and open to individual interpretation. Had the performance check been carried out using the system's performance monitor, and calibrated results recorded, system deterioration should have been recognised.

Figure 16

ROB F.O.		ROB DO.O.		ROB F.W.	
RADAR WATCHKEEPING PERFORMANCE					
WATCH	No. 1 / No. 2		PERFORMANCE CHECK RESULTS	TGT ACQUISITION PERFORMANCE	
	ON	OFF			
0000-0400					
0400-0800					
0800-1200	ON	ON	GOOD	GOOD	
1200-1600	ON	ON	GOOD	GOOD	
1600-2000	ON	ON	GOOD	GOOD	
2000-2400	ON	ON	GOOD	GOOD	
REMARKS FAILURES MAINTENANCE					

total radar
B/L = 2362.1
4656.5

Gas Monarch's radar performance check from Deck Log Book

Soon after the accident, the vessel's radars were inspected by service technicians in the port of Antwerp, where it was confirmed that the magnetron on each radar had deteriorated to well below optimum capability, and they were duly replaced. It was also found that the radar scanner bearings on the X band radar were worn to the extent that they allowed vertical play of 2cm in the antenna. A recommendation was made by the technicians to have this attended to as a matter of urgency.

Less than a year previously, the X band radar's magnetron had been replaced in Amsterdam. It is possible that the play in the radar scanner was contributory to the deterioration of the magnetron in such a short time span. The play in the scanner could also be responsible for deflecting some of the radar's energy from the wave guide and, thus, also contribute to poor performance.

In October 2006, the ship had an annual Cargo Ship Safety Equipment Certificate revalidated by LRS in Antwerp. The survey inspection for issue of this certificate includes such items as radar performance. However, as such vessels are frequently alongside gas terminals at the time of survey, it is not always possible to switch radars on; thus the surveyor will often rely on the master's or first officer's opinion of the equipment.

2.4 GAS MONARCH'S ACTIONS

2.4.1 Master

Gas Monarch's master joined the ship 3 weeks before the collision. He had been on the bridge with the 3/O a few days earlier, in fog in the Baltic, while the ship was passing through a fleet of fishing vessels. On that occasion, the 3/O had requested the master's presence on the bridge, as required by the master's standing orders. This gave the master the opportunity to observe his 3/O's ability and, from this, he felt confident that the 3/O would call him in other situations, if required.

From the changeover of the watch at 2000 hours, the master was on the bridge fairly regularly monitoring the watch, as well as attending to paperwork in his cabin. The master left the ship's bridge to go to bed approximately 40 minutes before the collision. At that time, visibility was severely restricted, the ship was sailing at full sea speed, no sound signals were being made and the ship was being steered by automatic pilot. This was in direct contravention of his personal standing orders, restricted visibility orders and night orders (**Annex 1**). The master's only verbal instruction to the 3/O was to call him if they encountered traffic.

Leaving the bridge in such conditions and navigational status could only have conveyed to the 3/O that the master's written orders were superficial and of little consequence. Additionally, the master's actions probably served to undermine the 3/O's training and instincts, thus preventing the 3/O from calling him when the radar contact with *Whispa* was lost.

It is surprising that an experienced master would act in this manner, not only in direct conflict with his own written orders, but also given the lack of experience of his watch keeping officer. The fact that the master did not recognise the potential outcome of his actions reflects a high degree of complacency on his part, a factor that MAIB sees as contributory in many accidents which it investigates.

On being called to the bridge after the accident, the master immediately stopped his ship, notified the coastguard, and stood by *Whispa* until the lifeboat arrived. However, he did not instruct his crew to make ready their rescue boat, which could have resulted in precious time being lost had *Whispa* required urgent assistance.

The master did not communicate directly with the skipper of *Whispa*. He felt that because he had identified his ship on radio, *Whispa*'s skipper would call him directly if he wished to communicate. Unfortunately, *Whispa*'s skipper had not realised from the communication with the coastguard that *Gas Monarch* was the colliding vessel but, rather, a random ship standing by to give assistance. It is required by international convention² and the UK Merchant Shipping Act 1995 that, following a collision, the vessels involved exchange pertinent information with each other.

Gas Monarch's crew were called to stations by use of the public address (PA) system rather than by the sounding of the general alarm (GA). The ship was then inspected for damage and stood by *Whispa* as requested by the coastguard.

² Article 8 of the 1910 Convention for the Unification of Certain Rules of Law with respect to Collisions between Vessels.

2.4.2 Third officer

It was the 3/O's first trip as a deck officer since his promotion from AB cadet. *Gas Monarch* was also the first ship he had sailed on without ARPA and due to his unfamiliarity with the electronic plotting function of the radars, preferred to rely on the AIS information displayed on the radar screen when considering developing traffic situations. *Whispa*, however, did not have AIS, and therefore was not displayed as an AIS contact, or plotted or systematically observed by *Gas Monarch*.

On taking over the watch at 2000, the 3/O sought clarification about the use of sound signals and was advised by the chief officer not to sound the horn unless traffic was encountered. His understanding of the word "traffic" was that it meant more than one vessel. On previous ships he had sailed on, the horn was used in restricted visibility, regardless of traffic density. The advice caused the 3/O concern, but he did not wish to question his superiors and, with reluctance, he carried on as instructed.

After detecting *Whispa* by radar at about 6-miles range, the 3/O quickly established the two vessels were on a collision course and altered *Gas Monarch*'s course 5° to port, from a course of 188° to 183°. Rule 8(b) requires *any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit, be large enough to be readily apparent to another vessel observing visually or by radar...* *Gas Monarch*'s 5° alteration of course was insufficient to have been apparent on *Whispa*'s radar screen.

Having executed the alteration to port, the 3/O then used a floating EBL to establish a CPA by simply aligning the EBL with the contact's trails. This use of the floating EBL to calculate *Whispa*'s CPA was unreliable due to the short trail length. An error of as little as 2° when projecting the trail over the length of the 6-mile range radar screen would have made a big difference to the indicated CPA. From observation of the floating EBL, the 3/O concluded the CPA of *Whispa* to be about 1 mile when the vessels were 5.5 miles apart; at 3 miles apart he revised the CPA to 7 cables (even though both vessels had, in reality, maintained a steady course). Post-collision examination of AIS and GPS data showed the CPA prior to *Whispa*'s alteration of course to be in the region of ⅓ of a mile (3.33 cables).

Rule 19 of the COLREGs requires, where a risk of collision has been determined by radar alone, vessels not in sight of one another in restricted visibility, should, *so far as possible*, avoid an alteration to port for vessels detected forward of the beam, and that any alteration is made in ample time. *Gas Monarch*'s 3/O made a port alteration because he felt a starboard adjustment would take the ship inshore towards shallow water. However, the nearest shallow water was some 3 miles to the west, thus giving ample room for *Gas Monarch* to alter course to starboard, in accordance with Rule 19. The chosen course of action, although made in ample time, was inappropriate and created a close quarters, starboard to starboard passing situation, with an unidentified vessel in dense fog.

Soon after the 3/O had estimated a CPA of 7 cables, *Whispa* was lost from both radars. Consequently he attempted to retune the radars by altering range scales and adjusting sea and rain clutter controls. On completion of adjustments, the radars were on auto tune, with sea anti-clutter set manually at about ⅓ sensitivity, while the rain anti-clutter was set at maximum suppression; these settings would have had the effect of masking weak images. Further confirmation of this is that, when on 1.5-mile range, the radar

screen was seen to clear with no speckles on it; a recognised sign of over-suppression. The combination of poor radar tuning due to operator inexperience, and magnetron deterioration, may explain the contact loss from the radars.

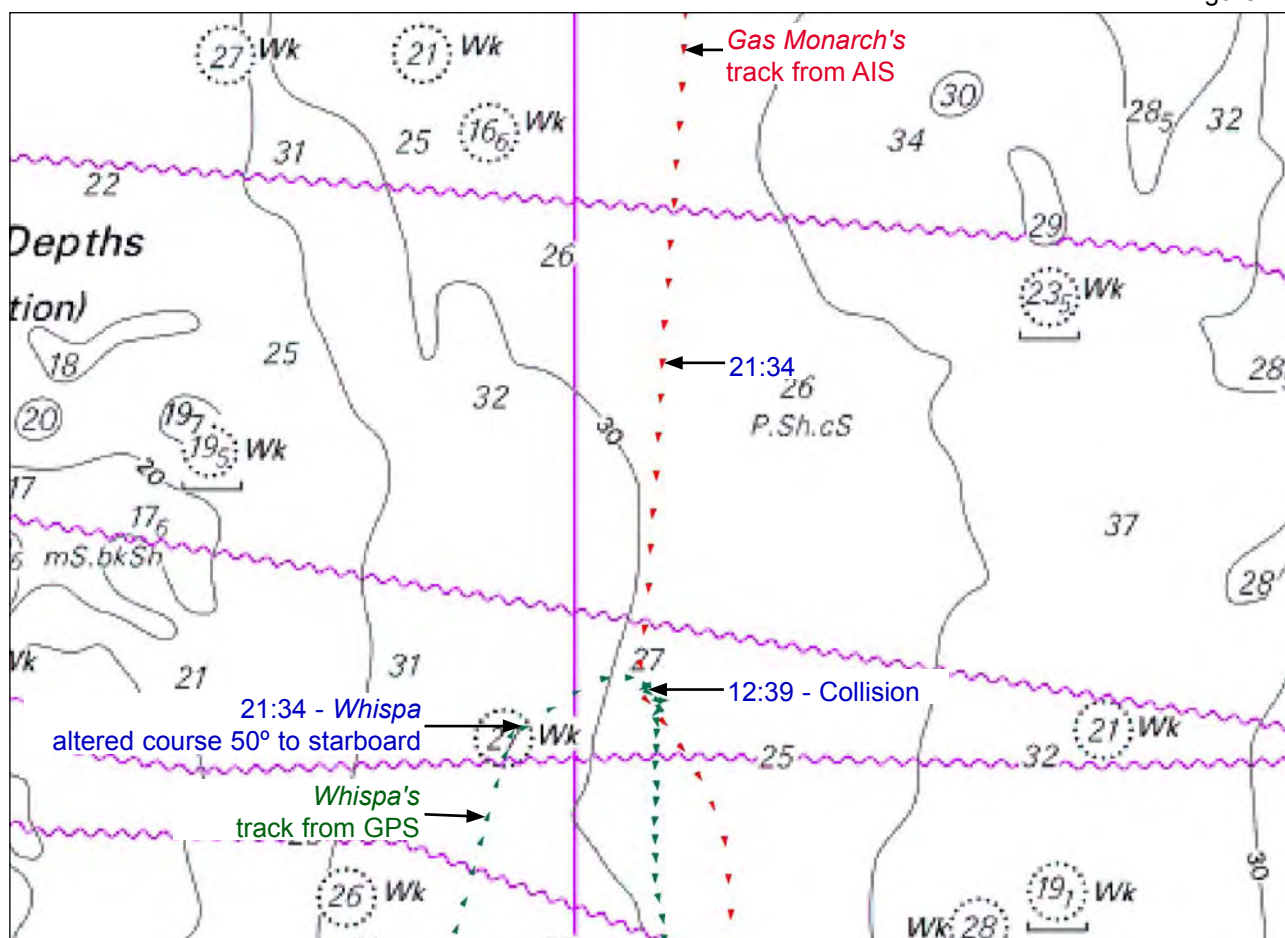
The 3/O's observations of *Whispa* were not systematic, as required by COLREGs Rule 7 (b), and the method of calculating CPA was unreliable. After losing the radar target he assumed that *Whispa* would pass clear on his starboard side and that there was no reason for *Gas Monarch* to alter course. Part (c) of Rule 7 stresses: *Assumptions shall not be made on the basis of scanty information, especially scanty radar information.* Once the radar target of *Whispa* was lost from *Gas Monarch*'s radar screens, it was inappropriate for the 3/O to allow *Gas Monarch* to continue on her course, unabated, on the assumption that *Whispa* would maintain her course and speed. *Gas Monarch*'s actions were based entirely on scanty radar information.

It would appear that *Gas Monarch*'s 3/O was confident that the lost contact was going to pass clear to starboard. This is evidenced by the 3/O going to the chart and fixing the vessel's position at the critical time of just 3 minutes before the collision. Nevertheless, the 3/O elected to occasionally shine his Aldis lamp into the fog as a precaution, just in case the lost radar contact did come too near. He recognised in hindsight that he should have used the fog horn, but elected the lamp, which made little impression in the fog. To not use sound signals in or near restricted visibility is a direct contravention of Rule 35 of the COLREGs.

When *Whispa*'s port navigation light appeared out of the fog, the 3/O immediately instructed the lookout to select hand-steering and to go hard starboard. At the same time, he shone the Aldis lamp towards the approaching vessel by way of a warning. As soon as the ship started to swing to starboard, a hard port instruction was given. These manoeuvres reduced the effect of the impact between the two vessels and helped to keep *Whispa* clear of *Gas Monarch*'s port side as they passed. The relative track movements of the two craft as they closed is shown clearly in **Figure 18** (note that *Gas Monarch*'s starboard alteration is not apparent on her AIS track due to its brevity, occurring between successive plots which were at 30 second intervals). The shining of the Aldis lamp, although well intentioned, destroyed *Whispa*'s skipper's night vision for a period of time and, again, the sound signal should have been used.

The 3/O ran to the port bridge wing in time to see the lights from *Whispa* disappear astern. He was not sure if *Gas Monarch* had collided with *Whispa*, but felt that, as he could see her lights, she must have been all right. He was stunned by the incident and stood, shaking, on the bridge for quite a while afterwards while *Gas Monarch* continued at full sea speed. It was not until he heard *Whispa*'s distress call approximately 8 minutes after the collision that he called the master to the bridge, whereupon the master took control.

Notwithstanding the 3/O may have been in a state of shock following the collision, it is of concern that his immediate reaction to the incident was not to take action to verify the status of *Whispa* and make preparations to provide the yacht's crew with assistance if required. There are very strong similarities between this accident and that of the loss of the sailing yacht *Ouzo* and her three crew (MAIB Report No 7/2007), where *Pride of Bilbao* came perilously close, or collided with the yacht, but carried on sailing as the lights of the yacht were still visible afterwards.



Relative tracks of *Gas Monarch* (reproduced from AIS) and of *Whispa* (reproduced from GPS)

2.4.3 Lookout, *Gas Monarch*

Gas Monarch's bridge team consisted of the 3/O and an AB acting as lookout. The bridge doors and windows were shut, reducing the possibility of anyone on the bridge hearing other vessels' sound signals. Additionally, in such restricted visibility, the ship's SMS required that she go to hand-steering and a helmsman be called to the bridge to allow the lookout to continue his duties.

Rule 5 of the COLREGs requires that "every vessel shall at all times maintain a proper look-out by sight and hearing ...". For the "hearing" part of this regulation to be effective, requires vessels to be making sound signals in restricted visibility, as required by Rule 35. Neither vessel was making such signals.

Gas Monarch had appropriate sound signalling equipment, with horns positioned forward and aft. Had she been sounding her horn, it is possible that it would have been detected on *Whispa*, thus giving an indication of proximity and relative position.

2.4.4 Speed, *Gas Monarch*

Rule 19 (b) also requires that in restricted visibility, "every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions."

It is appreciated that the criteria for determining a safe speed are open for debate and can be interpreted in different ways. However, a prudent approach would be to travel at a speed relative to the visibility, the characteristics of the vessel in question, the traffic density, navigational hazards and by having regard to the additional factors listed in Rule 6 (a) of the COLREGs.

At the time of the accident, *Gas Monarch* was sailing at full sea speed, before a tidal stream of some 2kts, giving a speed in excess of 14kts. The visibility from the time the master left the bridge was approximately 150m.

The use of radar has frequently, rightly or wrongly, encouraged seafarers to sail at speeds that would be totally unjustifiable without such an aid to navigation. Rule 6 (b) (i) recognises this fact and, in doing so, highlights the need for vessels with operational radar to be aware of the *characteristics, efficiency and limitations of the radar equipment*. *Gas Monarch's* 3/O was not aware of the radar's deteriorated efficiency when *Whispa's* contact disappeared from the radar. Once aware of his inability to regain the radar contact he should have reduced speed and called the master.

2.4.5 Commercial pressure

The master's decision to proceed at full sea speed in such conditions could have been due to commercial pressure. However, there was no suggestion of this from the master, and the managers of the vessel were content that this type of pressure was not applied by them or by the charterer.

The managers and owners do not condone their vessels travelling at an unsafe speed, at any time.

2.4.6 Bridge watchkeeping

The International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW) Chapter VIII lays down the regulatory standards for watchkeeping. Part 3 - "Watchkeeping at Sea" specifies the requirements that should be observed when maintaining a navigational watch. It is clear that these principles were not adhered to by the navigational team on *Gas Monarch*.

2.5 WHISPA'S ACTIONS

2.5.1 Skipper

Whispa's skipper had received no training in radar navigation and had little experience of using the equipment in fog. He monitored an unidentified vessel approach on radar, without applying the EBL, due to the contact being close to the heading marker. He was not aware of the target tracking facility on his radar, and thus did not utilise it as a means of recording the ship's earlier positions on his radar screen.

In their book "A Guide to the Collision Avoidance Rules," A N Cockroft and J N F Lameijer³ state; *Even continuous observation by a competent person is unlikely to be accepted as proper use of radar to obtain early warning of risk of collision if the bearings and distances of approaching vessels are not taken at regular intervals and carefully evaluated by plotting or by some equivalent method.*

It is very difficult to simply look at a contact's paint every few minutes and compare it to the previous estimated position from memory, or to be sure that the same part of the paint is being used for evaluation purposes. Using the EBL and target tracking facility on *Whispa's* radar might not have given definitive clarification, but it would certainly have been better than relying on positions from memory.

³ A N Cockroft and J N F Lameijer, "A Guide to the Collision Avoidance Rules." Sixth edition; NOV-2003; Butterworth-Heinemann; ISBN-13: 978-0-7506-6179-9; ISBN-10: 0-7506-6179-8

Had *Whispa*'s skipper been able to plot *Gas Monarch* effectively he would have seen that, although a close quarter's situation was developing, the two vessels were not on a collision course.

Cockroft and Lameijer go on to state in their publication: *Rule 8 (a) requires avoiding action to be taken in ample time in all conditions of visibility. When visibility is restricted it is generally necessary to take action to avoid a close quarters situation at an earlier stage. However, action should not be taken without first making a full assessment of the situation.*

Whispa's skipper made no attempt to estimate the speed of the closing contact, or the time to CPA. The method of determining the relative speed of an approaching radar contact was explained in the radar manual (**Annex 3**), which was stowed on board but the skipper was not aware of this information and his subsequent actions were taken without appropriate assimilation of the, albeit limited, information available.

Whispa's skipper was unsure about the configuration of his radar's range rings and the total distance he was observing ahead of his vessel created by the radar screen's oblong configuration (**Figures 14 and 15**). He was not aware that when on a 6-mile range scale the range rings were 1.5 miles apart, and when on 3-mile range the rings were 1 mile apart. Instead he believed them to be 1 mile and 0.5 mile apart respectively, as is more commonly found. However, regardless of the range scale in use, the skipper was confident that the helm alteration was demanded of the autopilot when the approaching vessel was close to the radar's 2.5 mile guard zone. Analysis of data stored by *Whispa*'s GPS showed that the alteration took effect when the vessels were 1.3 miles apart.

Whispa's autopilot was set to limit any helm applied to 4° to give a slow rate of turn and prevent over-steer in normal circumstances. The damping of the autopilot would have no doubt caused some delay to the actual course alteration taking effect, but it is questionable whether it would have delayed the turn by four minutes i.e. the time it would have taken for *Gas Monarch* to close from 2.5 miles to 1.3 miles. There was no magnetic compass at the navigation station, but prudent observation of the GPS would have shown that *Whispa*'s head was not swinging as demanded of the autopilot; there is no doubt that the delayed course alteration further endangered the yacht by placing her under the bows of the gas carrier.

Despite the skipper observing the approaching contact on the radar, the instrument was a potential liability rather than an aid, due to the faulty heading marker and his lack of familiarity with the equipment's functions.

While *Whispa*'s alteration of 50° to starboard was large enough to be readily apparent on a properly functioning radar, as Rule 8 (b) requires, a fundamental error of navigation occurred in the action not being taken in *ample time* as required by both Rules 8 and 19. At 1.3 miles apart, the vessels were some 4 minutes from collision. Even had the yacht skipper's perception of the distance between the two vessels been correct and the helm had responded instantly to his demand, the action to avoid the approaching close quarters situation would only have served to place *Whispa* at about the same CPA, but on the opposite side of *Gas Monarch*'s bow.

Rule 19 states that *an alteration to port [for vessels forward of the beam] should be avoided so far as possible.*

Whispa's skipper turned the vessel to starboard, hoping that the approaching ship was aware of the yacht's presence and would take the appropriate avoiding action of likewise altering her course to starboard. He was concerned that any alteration to port of *Whispa* would move the yacht towards the unidentified ship. The crucial point here for all seafarers is that, when presented with a developing close quarters situation, it must not be assumed that the other vessel is tracking your vessel's progress on radar; and emphasises the need to take avoiding action in *ample time*.

Rule 8 (e) states, "*if necessary to avoid collision or allow more time to assess the situation, a vessel shall slacken her speed or take all way off by stopping or reversing her means of propulsion.*" Although *Whispa* was travelling at a speed appropriate to the prevailing conditions, when her skipper became concerned about a situation developing, he could have gained more time to further assess the situation by slowing his vessel to the minimum speed required to maintain steerage. This would have made little difference to the CPA (had both vessels maintained their course), but it would have given the skipper more time to consider an alternative course of action. One option could have been to turn 180° onto a reciprocal course and run full speed down tide, thus gaining substantially more time to assess the situation.

2.5.2 Speed, *Whispa*

Whispa's speed over the ground of 4.5kts was not unreasonable for the conditions. Although Rule 6 of the COLREGs refers to Safe Speed, it is actually the only Rule that specifically makes reference to the *characteristics, efficiency and limitations of the radar equipment*. Rule 2 (a) however states that: (a) *Nothing in these Rules shall exonerate any vessel, or the owner, master or crew thereof, from the consequences of any neglect to comply with these Rules or of the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case.* Therefore the instructions regarding use of radar must be applied as a matter of good seamanship, and not only with safe speed in mind. *Whispa's* skipper had limited knowledge of the characteristics, efficiency and limitations of his radar equipment. Had he not had radar on board, it is doubtful he would have known of *Gas Monarch's* presence, and probably would have passed the other vessel without incident. The presence of the equipment encouraged *Whispa's* skipper to attempt to navigate by radar whilst unaware of the heading marker error and without having enough knowledge and understanding of the equipment's functionality to do so safely.

2.5.3 *Whispa's* radar reflectivity

Whispa's hull form was extremely smooth and streamlined, with an aluminium mast extending almost 20m above sea level. A radar reflector was suspended from the mast cross tree at about 6m above sea level (**Figure 19**). At no time previous to the accident had other skippers expressed concern to *Whispa's* skipper about the vessel's radar image.

Gas Monarch detected the yacht at some 6 miles distance, and continued to do so until the two vessels were just under 3 miles apart, when it disappeared from her screens; it has since been established that *Gas Monarch's* radar performance was deteriorated and that anti-rain clutter may have been set too high.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

The following safety issues have been identified as a result of the MAIB investigation. They are not presented in any order of priority.

3.2 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT WHICH HAVE RESULTED IN RECOMMENDATIONS

1. *Whispa's* radar had a heading marker error which was unrecognised by the skipper. [2.3.2]
2. *Whispa's* skipper had received no training in radar navigation and had little experience of using the equipment in fog. [2.5.1]
3. *Whispa's* skipper failed to appropriately apply the COLREGs. [2.5.1, 2.5.2]
4. There is disparity in the certification required by skipper SCV Code vessels. The BML certificate requires more comprehensive training, with particular regard to Radar and Fire Safety training, than RYA commercially endorsed certification. [2.6.2]

3.3 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE NOT RESULTED IN RECOMMENDATIONS BUT HAVE BEEN ADDRESSED

1. *Gas Monarch's* master left the bridge in the hands of his inexperienced 3/O in restricted visibility. [2.4.1]
2. *Gas Monarch's* master condoned the improper navigational status of his ship in restricted visibility. Doing so was a contradiction of COLREGs, STCW and EMS's Safety Management System, with particular respect to:
 - o Travelling at an unsafe speed for the prevailing weather conditions. [2.4.1]
 - o Restricted visibility sound signals were not used. [2.4.1]
 - o An extra hand was not on the bridge to allow hand-steering. [2.4.1, 2.4.3]
3. The actions of *Gas Monarch's* master not only indicated a high degree of complacency, but also set a bad example to the 3/O thus undermining the latter's training and ideal. [2.4.1]
4. *Gas Monarch's* 3/O lacked experience and familiarity with the ship's radar systems. [2.4.2]
5. The 3/O contravened COLREGs and STCW requirements in respect to keeping a safe navigational watch. [2.4.2, 2.4.3, 2.4.4]
6. The 3/O did not call the master when the radar contact was lost. [2.4.4]
7. Shining an Aldis lamp on *Whispa* destroyed her skipper's night vision for a period of time. [2.4.2]

8. *Gas Monarch*'s radar magnetrons had deteriorated beyond an acceptable level, with subsequent effects on the displayed image. [2.3.1]
9. *Gas Monarch*'s radar performance checks were not recorded appropriately. [2.3.1]
10. *Gas Monarch* had no ARPA radar which would have highlighted, by means of an alarm, *Whispa*'s disappearance from the radar screen. [2.3.1]
11. *Gas Monarch* failed to stop immediately after the collision. [2.4.2]
12. *Gas Monarch*'s crew were called to stations by PA system rather than the recognised GA. [2.4.1]
13. *Gas Monarch*'s rescue boat was not prepared to a state of readiness while standing by *Whispa*. [2.4.1]
14. The vessels failed to exchange information following the accident as required by international regulation. [2.4.1]
15. *Whispa*'s aft watertight bulkhead was breached and permitted through flooding. This went undetected due to the lack of early warning high level bilge alarms. [2.6.1]
16. *Whispa*'s survey and inspection under the SCV Code failed to detect the lack of high level bilge alarms, which endangered the craft when through flooding occurred. [2.6.1]